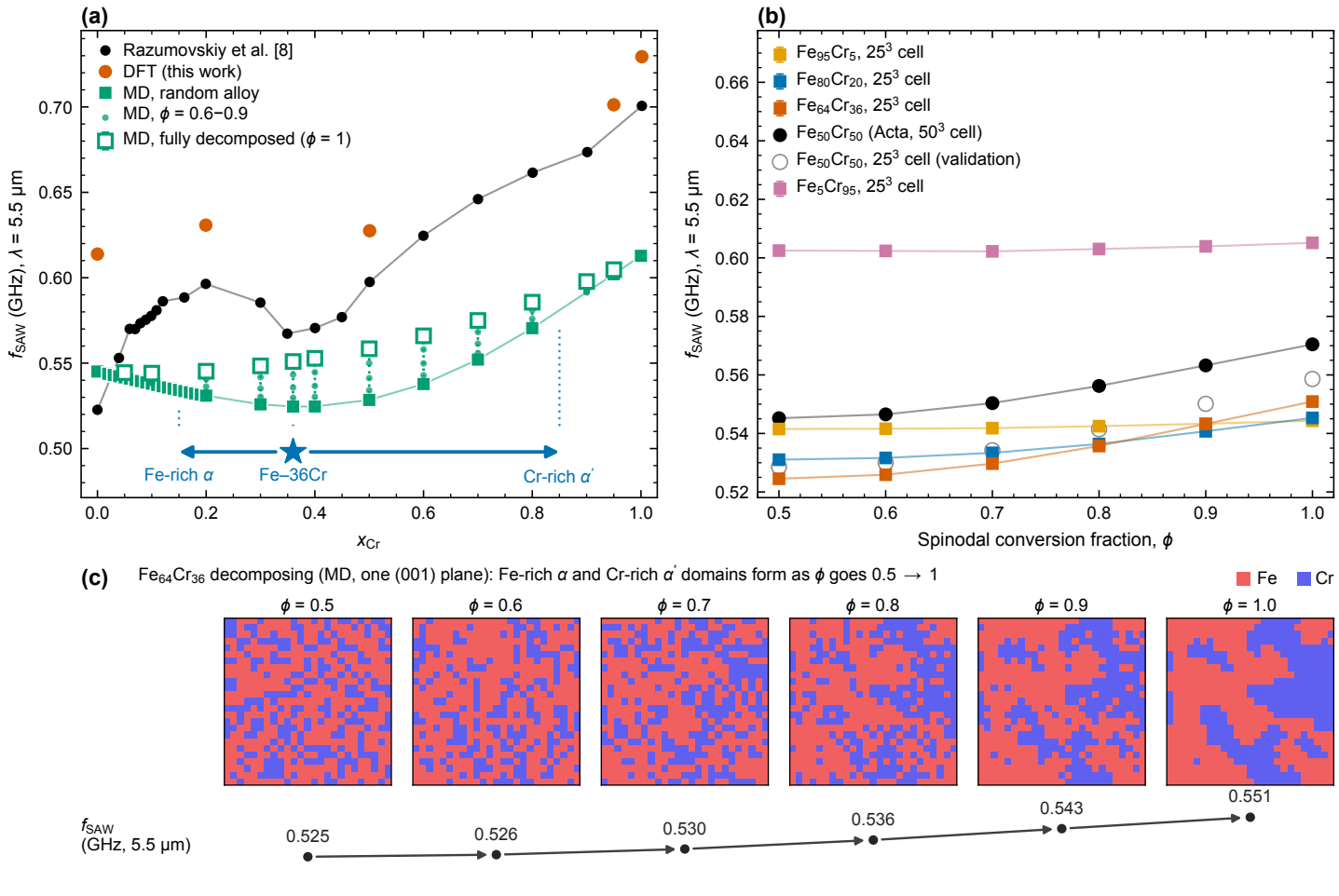

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, $\phi$ = 0.6–0.9
MD, fully decomposed ($\phi$ = 1)
Fe-rich $\alpha$
Fe–36Cr
Cr-rich $\alpha'$
$f_{SAW}$ (GHz), $\lambda$ = 5.5 μm
$x_{Cr}$
(b)
$Fe_{95}Cr_5$, $25^3$ cell
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$f_{SAW}$ (GHz), $\lambda$ = 5.5 μm
Spinodal conversion fraction, $\phi$
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, one (001) plane): Fe-rich $\alpha$ and Cr-rich $\alpha'$ domains form as $\phi$ goes 0.5 → 1
Fe
Cr
$\phi$ = 0.5
$\phi$ = 0.6
$\phi$ = 0.7
$\phi$ = 0.8
$\phi$ = 0.9
$\phi$ = 1.0
$f_{SAW}$ (GHz, 5.5 μm)
0.525
0.526
0.530
0.536
0.543
0.551